

Validation and method note

LUCKY CEMENT | MGT-353 | 5 October 2026

Company-Calibrated Synthetic Dataset for Academic Analysis — illustrative figures, calibrated to Lucky Cement's disclosed scale, not actual company records.

Scope, calibration and reproducibility

One illustrative domestic Karachi/Sindh dealer territory selling Ordinary Portland Cement is analysed in tonnes per week. A starting level of 1,550 tonnes and a modest 1.2-tonne-per-week underlying trend represent a small dealer territory rather than company-wide growth. Broad March-June construction strength adds 240 tonnes, while an assumed July-September monsoon block suppresses sales; these are teaching assumptions, not measured regional seasonality. Prices move in four steps from PKR 20,000 to PKR 22,800 per tonne, with a modest imperfect negative association; dealer discounts of 2%-5% occur in 34 weeks and produce approximately 14%-26% dispatch uplift, potentially including forward buying. A synthetic construction-activity index is chosen because project activity plausibly affects cement orders, and independent noise prevents a perfect relationship. Four one-off shocks and five stockout weeks create abnormal observed sales, while Gaussian noise of about 105 tonnes adds weekly variation. The data do not capture actual Lucky Cement sales, dealer credit, final-user consumption, brand mix, government project timing or true unconstrained demand.

FINANCIAL CALIBRATION

FY2026 standalone net revenue: PKR 136.527017 billion. Fiscal year ends June 30; sector cement. Reconciliation in PKR thousands: 192,861,184 gross revenue - 53,227,737 sales tax/FED - 3,106,430 rebates = 136,527,017 net revenue. Mirrored extracted statement rows were shifted; the company financial-performance summary and arithmetic were used, not corrupted row labels. FY2026 synthetic territory net-sales proxy: PKR 2.021 billion, 1.48% of standalone revenue. This is a scale sanity check, not a verified territory share. Group revenue is not used to calibrate a single cement territory. The latest annual report is recorded by PSX but its PDF could not be fetched in this environment; the financial results summary is the accessible calibration evidence.

Seed 3532026. Dataset SHA-256: a12f06c4579e96303885bb629bb9717b8e7fe90112580a31950d89a132171269. File values are frozen and reused. Student identity is taken from prior academic work and should be confirmed before submission.

Evidence and source limitations

Lucky Cement / PSX: FY2026 financial results, 10 August 2026; annual report transmission 4 September 2026
<https://dps.psx.com.pk/company/LUCK>

Lucky Cement: Financial reports landing page, FY2026 available
<https://lucky-cement.com/en/corporate/investor-relations/financial-reports>

Company-authored disclosure, mirrored transcription: FY2026 unconsolidated financial-performance summary; net revenue PKR 136,527m
<https://financialfilings.com/filings/lucky-cement-limited/annual-report/2026/55070674/>

Lucky Cement: Product portfolio context: OPC, SRC and composite cement
<https://lucky-cement.com/en>

Course assignment: Portfolio Milestone 1(1).pdf, Sections 4.2, 6-17
Teacher-provided attachment

Dataset dictionary

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Field / unit	Definition and source logic
Dataset_Disclosure Text	Mandatory synthetic-data disclosure Disclosure. Repeated in each record so exported data retain their classification
Week_Start_Date Date	Monday of sales week Synthetic structure. 156 consecutive weeks; 2023-10-09 to 2026-09-28
Product_or_Category Text	Ordinary Portland Cement Real category / assumed scope. Company portfolio includes OPC; no specific brand mix claimed
Location_or_Channel Text	Illustrative Karachi/Sindh dealer territory Assumed. A territory, not whole-company sales
Actual_Demand_Tonnes Tonnes/week	Observed dispatch/sales target Synthetic. Base 1,550 t/wk with drivers/noise; constrained in stockout weeks
Historical_Forecast_Tonnes Tonnes/week	Illustrative prior weekly forecast Synthetic. Prior four-week mean plus 2% bias and forecast noise; first week 1,550 baseline
Price_PKR_per_Tonne PKR/tonne	Undiscounted net selling-price proxy Assumed/synthetic. 20,000 to 22,800 in four steps; excludes indirect taxes
Promo_Flag 0/1	Dealer incentive active Synthetic. 34 of 156 weeks, selected independently of season
Promo_Depth Percentage points	Discount during incentive Synthetic. 2%-5% when active; zero otherwise
Event_Flag 0/1	Broad construction season Assumed calendar rule. 1 for March-June; not a Ramadan/Eid flag
Construction_Activity_Index Index points	Territory project-activity proxy Synthetic. Around 100; modest trend, seasonal component and independent noise
Monsoon_Flag 0/1	Illustrative slower building season Assumed calendar rule. 1 for July-September; not recorded weather observations
Stockout_Flag 0/1	Supply/dispatch constrained sales Synthetic. Five weeks; sales do not reveal unconstrained demand
Exceptional_Event Text	One-off incident description Synthetic. Four incidents, each one or two weeks; blank otherwise

Holdout design and accuracy

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Training: 9 October 2023-6 July 2026, 144 weeks. Holdout: 13 July-28 September 2026, 12 weeks. The five specifications were fixed before evaluation. MA predictions are appended recursively; holdout actuals never replace them. Seasonal-trend uses only training demand and known calendar dates. Full-history statistics are descriptive and separately labelled. Regression uses training rows only. Forward forecasts refit the same selected method using all 156 observations after evaluation.

Method	MAD t	MAPE %	WAPE %	Bias t	TS
Calendar seasonal trend	105.8	8.27	7.71	-36.3	-4.12
Naive	120.7	9.55	8.79	-60.7	-6.03
Seasonal naive (52 weeks)	131.9	9.66	9.61	51.6	4.69
Recursive MA (4 weeks)	360.0	27.70	26.23	-360.0	-12.00
SES (alpha=0.3)	416.7	31.89	30.36	-416.7	-12.00

Winner RMSE 134.49 t; RSFE -435.89 t. Positive Actual-Forecast means underforecast; negative means overforecast. MAPE is safe for these positive observations but unstable with zero/near-zero actuals. WAPE weights absolute error by total volume; RMSE highlights large misses. TS -4.12 already exceeds the assumed +/-4 control band. The plan is provisional. There is no perfect accuracy; leakage checks passed for target values. No claim is made that choosing the winner on one holdout gives an unbiased future error estimate.

Date	Actual t	Forecast t	Error t	Diagnosis
2026-08-17	1041	1381.3	-340.3	Availability-constrained observed sales; date-only model cannot anticipate unannounced dispatch interruption.
2026-07-27	1520	1374.5	145.5	Dealer promotion associated with uplift; seasonal model does not explicitly model planned discounts.
2026-08-03	1237	1379.0	-142.0	Monsoon-period variability, construction-activity variation and noise; a calendar effect does not explain every week.
2026-07-20	1498	1373.4	124.6	Monsoon-period variability, construction-activity variation and noise; a calendar effect does not explain every week.
2026-09-28	1587	1471.0	116.0	Monsoon-period variability, construction-activity variation and noise; a calendar effect does not explain every week.

Worked metric verification

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All 12 held-out weeks for the winning model. Display rounding is for readability; calculations retain full precision. Reproduce MAD, WAPE and Bias in Excel/by hand as your independent verification. The code already cross-checks all seven metrics using a separate scalar loop.

Week	Actual	Forecast	A - F	A - F
2026-07-13	1329	1372.279	-43.279	43.279
2026-07-20	1498	1373.405	124.595	124.595
2026-07-27	1520	1374.532	145.468	145.468
2026-08-03	1237	1379.034	-142.034	142.034
2026-08-10	1289	1380.160	-91.160	91.160
2026-08-17	1041	1381.286	-340.286	340.286
2026-08-24	1394	1382.413	11.587	11.587
2026-08-31	1319	1383.539	-64.539	64.539
2026-09-07	1487	1467.622	19.378	19.378
2026-09-14	1357	1468.748	-111.748	111.748
2026-09-21	1410	1469.875	-59.875	59.875
2026-09-28	1587	1471.001	115.999	115.999

Actual total = 16468.000 t; absolute-error total = 1269.949 t; signed-error total = -435.894 t.

MAD = $1269.949 / 12 = 105.829$ t.

WAPE = $1269.949 / 16468.000 \times 100 = 7.712\%$.

Bias = $-435.894 / 12 = -36.325$ t; RSFE = -435.894 t.

Tracking Signal = $-435.894 / 105.829 = -4.119$.

RMSE = square root ($217040.123 / 12$) = 134.487 t.

MAPE = $(0.992050 / 12) \times 100 = 8.267\%$.

Regression: association, not proof

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OLS uses 144 training observations and seven explanatory variables plus intercept. R-squared 0.8567; adjusted R-squared 0.8493. Degrees of freedom 136. Standard OLS p-values are reported as statistical learning outputs. Durbin-Watson 1.832; scaled design condition number 264.8, warning of strong collinearity. Diagnostics do not establish valid causal identification.

Variable	Coefficient	p-value	Business interpretation
Intercept	1143.608	0.19	Outside observed zero-price/zero-index range; not operational baseline.
Price_PKR_per_Tonne	-0.047	0.275	+PKR 1,000/t associated with -46.9 t/week; weak statistical evidence.
Promo_Flag	371.769	5.1e-31	0 to 1 dealer incentive, holding other included drivers constant.
Event_Flag	224.607	1.01e-15	0 to 1 March-June season indicator.
Construction_Activity_Index	13.426	3.74e-09	Per index point, holding other drivers constant.
Monsoon_Flag	-324.463	9.45e-17	0 to 1 July-September indicator.
Stockout_Flag	-584.448	7.11e-16	Availability-constrained sales association, not lost true demand estimate.
Time_Weeks	1.647	0.0875	Per additional week controlling included variables.

Coefficient signs are directionally plausible: incentives, construction season and activity positive; monsoon and stockout negative. Price is negative but imprecise due in part to step prices and time trend. No pricing policy follows from this estimate. One potential confounder is dealer stock accumulation: promotions may pull orders forward rather than increase final consumption. Credit terms, competitor pricing, project starts and government spending are omitted. Ordinary p-values assume independent homoskedastic residuals; a more advanced analysis would use time-series-robust uncertainty. Synthetic generation embeds the relationships, so significance is not independent real-company evidence.

Managerial implication within this illustration: prepare dispatch flexibility for incentive weeks; verify whether uplift is incremental or purchases shifted in time. Regression is explanatory, not a separately validated future-demand model; no held-out realised construction index or stockout flag enters the selected time-series forecast.

Stress test and release conditions

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Future total 22,109.0 t; commitment 17,687.2 t; flexible balance 4,421.8 t. Exact split reconciles to the forecast. Buffer 211.7 t is separate from expected demand. Each week, responsive supply is $\min(\text{capacity}, \max(80\% \text{ of base forecast, scenario demand}))$; opening buffer can cover deficits. Shortfalls are lost sales, not later backorders. Future band ± 2 MAD is uncalibrated. Model specifications remain unchanged when refitting all history.

Scenario	Demand t	Fixed excess t	Fixed short t	Responsive short t
Base	22109	0	0	0
Downside -20%	17687	4422	0	0
Upside +20%	26531	0	4422	0
Dealer event +30% first 2 weeks	23235	0	1126	0
Dispatch capacity -40% first 2 weeks	22109	0	872	661

Financial assumptions: inventory valuation at PKR 17,000/t and lost contribution at PKR 4,500/t. Inventory value is cash tied up, not an immediate expense or forecast profit loss. Lost contribution is estimated margin at risk, not lost revenue. Financing, storage, tax and spoilage costs are excluded. Flexible response and dispatch capacity must be verified before a real release. A forecast can be accurate while a dispatch interruption still makes its operating plan fail.

Use the forecast provisionally: the holdout Tracking Signal already breaches the assumed ± 4 band, requiring immediate stockout/bias review before operational release. Schedule 17,687 tonnes (80%) across 12 weeks; retain 4,422 tonnes (20%) of base-volume flexibility and reserve access to 20% upside capacity. Hold an illustrative 212-tonne buffer separately, costing PKR 3.60m under the assumed unit cost. Do not prebuild all 12 weeks of cement. Reforecast weekly; escalate if either of the first two actual weeks differs from forecast by more than 10%, rolling Tracking Signal exceeds ± 4 , or a dispatch interruption is announced. The ± 4 band is an assumed monitoring rule, not a proven service guarantee.

Audit status and remaining student actions

PASS: date ordering, no gaps, frozen data, common holdout, no target leakage, recursive MA, separate training regression, scalar metric reconciliation, five-miss diagnosis, four stress conditions, forward-refit separation.

CAUTION: current bias control breach; one short seasonal holdout; synthetic drivers and costs; full annual-report PDF inaccessible. STUDENT: independently reproduce three metrics, review/explain outputs, confirm name and assumptions, then submit files on LMS and GitHub. Uploads to either destination have not been performed.